

Trajectory Uncertainty Calibration for Multimodal VLMs

Complete Benchmark & Transfer LaTeX Table Compendium

M3-LLaVA (7B) & MQT-LLaVA (7B) Evaluations

September 30, 2026

Abstract

This document compiles all 37 publication-ready LaTeX tables across Groups 1–8 and RQ1 Master Tables for local inspection before Overleaf integration. All tables follow the standardized 4-metric schema: ECE (%) ↓, Ada-ECE (%) ↓, Brier score ↓, and AUROC ↑, featuring Option A ranking (**bold** Rank 1, *italic* Rank 2), Core 5 macro aggregation, Trajectory Platt (5D) evaluation, dedicated VQAv2 single-pass vs. multi-rollout comparison, and isolated RQ1 Core 5 and Adversarial master benchmarks.

Contents

1	Main Manuscript Tables (Groups 5, 2, 6, 7, 4, 9)	2
1.1	Table 2: Zero-Shot Cross-Domain Calibration Transfer (M3-LLaVA 7B)	2
1.2	Table 3: Zero-Shot Cross-Domain Calibration Transfer (MQT-LLaVA 7B)	3
1.3	Group 2: Macro-Averaged Calibration Benchmark Across Core 5 Datasets	4
1.4	Comprehensive Core 5 Dataset Benchmark Breakdown (M3 & MQT 7B)	5
1.5	RQ1 Master Table: Core 5 Vision-Language Calibration Benchmark (M3 & MQT 7B) . .	6
1.6	RQ1 Master Table: Adversarial & Safety Benchmark (M3 & MQT 7B)	7
1.7	Group 9: VQAv2 Single-Pass (1×) vs. Multi-Rollout UMPIRE Suite (10×)	8
1.8	Group 6: Zero-Shot Source Anchor Leaderboard	9
1.9	Group 7: Sample Efficiency & Low-Data Calibration Scaling	10
1.10	Group 4: Leave-One-Dataset-Out (LODO) Cross-Domain Transfer	13
2	Group 3: Temperature Transfer Robustness Ablation	14
2.1	Temperature Transfer: Macro-Averaged Across 4 Benchmarks	14
2.2	Temperature Transfer: POPE	17
2.3	Temperature Transfer: ScienceQA	20
2.4	Temperature Transfer: TextVQA	23
2.5	Temperature Transfer: VizWiz-VQA	26
3	Group 1: Per-Dataset In-Domain Calibration Benchmarks	29
3.1	Benchmark: AI2D	29
3.2	Benchmark: Adversarial VQA	30
3.3	Benchmark: CHARTQA	31
3.4	Benchmark: DOCVQA	32
3.5	Benchmark: GQA	33
3.6	Benchmark: INFOGRAPHICVQA	34
3.7	Benchmark: LEGOPUZZLES	35
3.8	Benchmark: MMBENCH	36
3.9	Benchmark: MMMU	37
3.10	Benchmark: POPE	38
3.11	Benchmark: SCIENCEQA	39
3.12	Benchmark: SEEDBENCH	40
3.13	Benchmark: TEXTVQA	41

3.14	Benchmark: VIZWIZVQA	42
3.15	Benchmark: VLLM-SAFETY	43
3.16	Benchmark: VQAV2	44
4	Group 8: Combinatorial 5D Trajectory Feature Ablation Study	45
4.1	Table 4: Macro Cardinality Progression ($k \in \{1 \dots 5\}$)	45
4.2	Table 5: Leave-One-Out (LOO) Feature Degradation Sensitivity	46
4.3	Table 6: Exhaustive 14-Benchmark Adaptive ECE Progression Grid	47
4.4	Table 7: Cardinality Progression on TextVQA	48
4.5	Table 8: Cardinality Progression on VQAv2	49
4.6	Table 9: Cardinality Progression on LegoPuzzles	50

Source: VQAv2 (General Scene VQA)												
Method	TextVQA (Scene Text)				AI2D (Diagrams)				Adversarial VQA (Adversarial Robustness)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0578	0.0535	0.835	0.1628	0.0861	0.1115	0.661	0.2289	0.0526	0.0745	0.691	0.2240
Uncalibrated (NC)	<i>0.0851</i>	<i>0.0867</i>	<i>0.837</i>	0.1717	0.4210	0.4210	0.638	0.4209	0.2370	0.2335	0.594	0.3219
Temperature Scaling (TS)	0.1030	0.1012	<i>0.837</i>	0.1739	0.4201	0.4201	0.638	0.4200	0.2047	<i>0.2102</i>	0.594	0.3065
Adaptive TS (ATS)	0.1154	0.1126	0.840	<i>0.1737</i>	0.4196	0.4196	0.667	0.4194	<i>0.2058</i>	0.2079	<i>0.604</i>	<i>0.3020</i>
Platt Scaling (1D)	0.1788	0.1785	<i>0.837</i>	0.2066	<i>0.4184</i>	<i>0.4184</i>	0.638	<i>0.4183</i>	0.2791	0.2791	0.594	0.3257
Platt 5D (Ours)	0.0736	0.0754	0.811	0.1793	0.4026	0.4026	<i>0.661</i>	0.4027	0.2409	0.2402	0.639	0.2986
Source: VizWiz-VQA (Assistive / Blurry)												
Method	TextVQA (Scene Text)				AI2D (Diagrams)				VQAv2 (General Scene VQA)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0578	0.0535	0.835	0.1628	0.0861	0.1115	0.661	0.2289	0.0686	0.0650	0.788	0.1380
Uncalibrated (NC)	0.0851	0.0867	0.837	0.1717	0.4210	0.4210	0.638	0.4209	0.0791	0.0769	<i>0.764</i>	0.1370
Temperature Scaling (TS)	0.0886	<i>0.0851</i>	0.837	0.1717	0.4207	0.4207	0.638	0.4206	<i>0.0892</i>	<i>0.0915</i>	<i>0.764</i>	<i>0.1407</i>
Adaptive TS (ATS)	0.0853	0.0854	<i>0.835</i>	0.1720	0.4207	0.4207	0.605	<i>0.4207</i>	0.0915	0.0916	0.760	0.1414
Platt Scaling (1D)	<i>0.0842</i>	0.0896	0.837	<i>0.1689</i>	<i>0.4209</i>	<i>0.4209</i>	0.638	0.4208	0.1779	0.1782	<i>0.764</i>	0.1767
Platt 5D (Ours)	0.0776	0.0779	<i>0.835</i>	0.1683	<i>0.4209</i>	<i>0.4209</i>	<i>0.637</i>	0.4208	0.1788	0.1776	0.765	0.1772
Source: TextVQA (Scene Text)												
Method	VQAv2 (General Scene VQA)				Adversarial VQA (Adversarial Robustness)				VLLM-Safety (Adversarial Safety)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0686	0.0650	0.788	0.1380	0.0526	0.0745	0.691	0.2240	0.0587	0.0688	0.773	0.1709
Uncalibrated (NC)	0.0791	<i>0.0769</i>	0.764	0.1370	0.2370	0.2335	0.594	0.3219	<i>0.1157</i>	<i>0.1119</i>	0.821	<i>0.1763</i>
Temperature Scaling (TS)	0.0811	0.0799	0.764	0.1380	0.2290	0.2270	0.594	0.3185	0.1185	0.1164	0.821	0.1757
Adaptive TS (ATS)	<i>0.0799</i>	0.0757	<i>0.768</i>	<i>0.1372</i>	0.2327	0.2296	<i>0.597</i>	0.3187	0.1216	0.1169	<i>0.820</i>	0.1770
Platt Scaling (1D)	0.1235	0.1241	0.764	0.1507	0.2227	0.2162	0.594	<i>0.3177</i>	0.1223	0.1203	0.821	0.1836
Platt 5D (Ours)	0.0903	0.0928	0.770	0.1417	<i>0.2251</i>	<i>0.2245</i>	0.599	0.3162	0.1013	0.0973	0.812	0.1774

Table 1: **Zero-Shot Cross-Domain Calibration Transfer (M3-LLaVA 7B)**. Pairwise zero-shot transfer across held-out target benchmarks without target supervision. Evaluated on ECE (↓), Ada-ECE (↓), AUROC (↑), and Brier score (↓). **Bold**: Rank 1; *italic*: Rank 2 among transfer methods.

1 Main Manuscript Tables (Groups 5, 2, 6, 7, 4, 9)

1.1 Table 2: Zero-Shot Cross-Domain Calibration Transfer (M3-LLaVA 7B)

Source: VQAv2 (General Scene VQA)												
Method	TextVQA (Scene Text)				AI2D (Diagrams)				Adversarial VQA (Adversarial Robustness)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0525	0.0776	0.719	0.1948	0.0627	0.0653	0.739	0.2035	0.0502	0.0654	0.654	0.2332
Uncalibrated (NC)	0.3168	0.3168	0.722	0.3096	0.2631	0.2625	0.733	0.2909	0.2726	0.2716	0.591	0.3388
Temperature Scaling (TS)	<i>0.2757</i>	<i>0.2757</i>	0.722	0.2753	0.1745	0.1743	0.733	0.2387	0.2191	<i>0.2240</i>	0.591	0.3112
Adaptive TS (ATS)	0.2758	0.2758	0.740	<i>0.2705</i>	<i>0.1818</i>	<i>0.1818</i>	<i>0.729</i>	<i>0.2423</i>	<i>0.2206</i>	0.2219	<i>0.601</i>	<i>0.3070</i>
Platt Scaling (1D)	0.3714	0.3714	0.722	0.3388	0.2362	0.2362	0.733	0.2718	0.2946	0.2946	0.591	0.3349
Platt 5D (Ours)	0.2356	0.2357	<i>0.730</i>	0.2569	0.2130	0.2112	0.648	0.2761	0.2331	0.2293	0.664	0.2858
Source: VizWiz-VQA (Assistive / Blurry)												
Method	TextVQA (Scene Text)				AI2D (Diagrams)				VQAv2 (General Scene VQA)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0525	0.0776	0.719	0.1948	0.0627	0.0653	0.739	0.2035	0.0716	0.0764	0.739	0.1629
Uncalibrated (NC)	0.3168	0.3168	<i>0.722</i>	0.3096	0.2631	0.2625	<i>0.733</i>	0.2909	0.0745	0.0748	<i>0.720</i>	0.1568
Temperature Scaling (TS)	0.2565	0.2565	<i>0.722</i>	0.2654	0.1424	0.1415	<i>0.733</i>	0.2283	<i>0.1371</i>	<i>0.1381</i>	<i>0.720</i>	<i>0.1727</i>
Adaptive TS (ATS)	0.2588	0.2588	0.714	0.2686	0.1448	0.1438	0.735	0.2287	0.1486	0.1444	0.715	0.1739
Platt Scaling (1D)	<i>0.1475</i>	<i>0.1447</i>	<i>0.722</i>	<i>0.2331</i>	<i>0.1121</i>	<i>0.1131</i>	<i>0.733</i>	0.2190	0.3261	0.3266	<i>0.720</i>	0.2837
Platt 5D (Ours)	0.1340	0.1316	0.736	0.2185	0.1026	0.1058	0.727	<i>0.2219</i>	0.3148	0.3144	0.730	0.2724
Source: TextVQA (Scene Text)												
Method	VQAv2 (General Scene VQA)				Adversarial VQA (Adversarial Robustness)				VLLM-Safety (Adversarial Safety)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0716	0.0764	0.739	0.1629	0.0502	0.0654	0.654	0.2332	0.0568	0.0658	0.733	0.1921
Uncalibrated (NC)	0.0745	0.0748	0.720	0.1568	0.2726	0.2716	0.591	0.3388	<i>0.0851</i>	<i>0.0787</i>	0.772	0.1970
Temperature Scaling (TS)	0.2230	0.2409	0.720	0.2190	0.0813	0.1347	0.591	<i>0.2534</i>	0.1744	0.2078	0.772	0.2353
Adaptive TS (ATS)	<i>0.2200</i>	<i>0.2374</i>	<i>0.723</i>	<i>0.2167</i>	<i>0.0892</i>	<i>0.1342</i>	<i>0.593</i>	0.2537	0.1793	0.2076	0.772	0.2350
Platt Scaling (1D)	0.3649	0.3632	0.720	0.3048	0.2030	0.1987	0.591	0.2780	0.1188	0.1341	0.772	0.2052
Platt 5D (Ours)	0.2881	0.2881	0.755	0.2402	0.1061	0.1015	0.626	0.2481	0.0622	0.0651	<i>0.721</i>	<i>0.1987</i>

Table 2: **Zero-Shot Cross-Domain Calibration Transfer (MQT-LLaVA 7B)**. Pairwise zero-shot transfer across held-out target benchmarks without target supervision. Evaluated on ECE (↓), Ada-ECE (↓), AUROC (↑), and Brier score (↓). **Bold**: Rank 1; *italic*: Rank 2 among transfer methods.

1.2 Table 3: Zero-Shot Cross-Domain Calibration Transfer (MQT-LLaVA 7B)

Calibration Method	Regime / Sampling	Macro ECE ↓	Macro Ada-ECE ↓	Macro Brier ↓	Macro AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.1985	0.1964	0.2348	0.765
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.1132	0.1194	0.1919	0.765
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.1101	0.1197	0.1909	<i>0.771</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.0621	0.0673	<i>0.1788</i>	0.765
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>0.0656</i>	<i>0.0715</i>	0.1769	0.778

Table 3: **Macro-Averaged Calibration Benchmark Across Core 5 Datasets (M3-LLaVA 7B).**

Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	Macro ECE ↓	Macro Ada-ECE ↓	Macro Brier ↓	Macro AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.2484	0.2460	0.2580	0.726
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.1682	0.1694	0.2158	0.726
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.1639	0.1659	0.2145	<i>0.730</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>0.0793</i>	<i>0.0884</i>	<i>0.1845</i>	0.726
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.0624	0.0716	0.1778	0.752

Table 4: **Macro-Averaged Calibration Benchmark Across Core 5 Datasets (MQT-LLaVA 7B).**

Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

1.3 Group 2: Macro-Averaged Calibration Benchmark Across Core 5 Datasets

Calibration Method	AI2D		ScienceQA		TextVQA		VizWiz-VQA		VQAv2		Average	
	Ada ↓	AUC ↑	Ada ↓	AUC ↑	Ada ↓	AUC ↑	Ada ↓	AUC ↑	Ada ↓	AUC ↑	Ada ↓	AUC ↑
Panel A: M3-LLaVA (7B)												
Naive Confidence (NC)	0.4200	0.614	0.2193	0.807	0.1107	0.839	0.1597	<i>0.793</i>	0.0723	<i>0.772</i>	0.1964	0.765
Temperature Scaling (TS)	0.1218	0.614	0.1268	0.807	0.1114	0.839	0.1509	<i>0.793</i>	0.0858	<i>0.772</i>	0.1194	0.765
Adaptive TS (ATS)	0.1315	<i>0.625</i>	0.1300	<i>0.808</i>	0.1131	0.839	0.1487	<i>0.793</i>	0.0752	0.788	0.1197	<i>0.771</i>
Platt Scaling (1D)	0.0770	0.614	0.0499	0.807	<i>0.0822</i>	0.839	0.0580	<i>0.793</i>	<i>0.0697</i>	<i>0.772</i>	0.0673	0.765
Trajectory Platt (5D)	<i>0.1115</i>	0.661	<i>0.0615</i>	0.813	0.0535	<i>0.835</i>	<i>0.0657</i>	0.794	0.0650	0.788	<i>0.0715</i>	0.778
Panel B: MQT-LLaVA (7B)												
Naive Confidence (NC)	0.2692	<i>0.746</i>	0.2317	0.760	0.3191	0.684	0.3066	0.797	<i>0.1031</i>	0.641	0.2460	0.726
Temperature Scaling (TS)	0.1310	<i>0.746</i>	0.1023	0.760	0.1832	0.684	0.2941	0.797	0.1365	0.641	0.1694	0.726
Adaptive TS (ATS)	0.1408	0.747	0.0938	<i>0.761</i>	0.1836	<i>0.689</i>	0.2911	<i>0.796</i>	0.1203	<i>0.658</i>	0.1659	<i>0.730</i>
Platt Scaling (1D)	<i>0.0825</i>	<i>0.746</i>	<i>0.0691</i>	0.760	<i>0.1046</i>	0.684	0.0762	0.797	0.1094	0.641	<i>0.0884</i>	0.726
Trajectory Platt (5D)	0.0653	0.739	0.0572	0.769	0.0776	0.719	<i>0.0817</i>	0.795	0.0764	0.739	0.0716	0.752

Table 5: **Calibration Benchmark Breakdown Across Core 5 Datasets.** Evaluated on Ada-ECE (↓) and AUROC (↑) for M3-LLaVA (7B) (Panel A) and MQT-LLaVA (7B) (Panel B). **Bold:** best; *italic:* second best within each metric column.

1.4 Comprehensive Core 5 Dataset Benchmark Breakdown (M3 & MQT 7B)

Dataset	Naive Confidence			Temperature Scaling			Adaptive TS (ATS)			Platt Scaling (1D)			Trajectory Platt (5D)		
	ECE	Ada	Brier AUC	ECE	Ada	Brier AUC	ECE	Ada	Brier AUC	ECE	Ada	Brier AUC	ECE	Ada	Brier AUC
AI2D	0.420	0.420	0.420	0.614	0.086	0.121	0.247	0.625	0.056	0.077	0.237	0.614	0.086	0.111	0.228
ScienceQA	0.221	0.219	0.236	0.807	0.131	0.126	0.192	0.807	0.129	0.130	0.191	0.808	0.048	0.049	0.173
TextVQA	0.105	0.110	0.171	0.839	0.103	0.111	0.171	0.839	0.101	0.113	0.171	0.839	0.084	0.082	0.161
VizWiz-VQA	0.159	0.159	0.207	0.793	0.150	0.150	0.206	0.793	0.148	0.148	0.205	0.793	0.066	0.058	0.182
VQAv2	0.086	0.072	0.138	0.772	0.094	0.085	0.141	0.772	0.083	0.075	0.138	0.788	0.055	0.069	0.139
Macro Avg.	0.198	0.196	0.234	0.765	0.113	0.119	0.191	0.765	0.110	0.119	0.190	0.771	0.062	0.067	0.178

Table 6: **RQ1 Calibration Benchmark on M3-LLaVA (7B)**. Comparison of calibration methods across 5 vision-language benchmarks. Metrics: Expected Calibration Error (ECE ↓), Adaptive ECE (Ada-ECE ↓), Brier Score (↓), and AUC (↑). **Bold**: Rank 1; *italic*: Rank 2 among single-pass methods.

Dataset	Naive Confidence			Temperature Scaling			Adaptive TS (ATS)			Platt Scaling (1D)			Trajectory Platt (5D)		
	ECE	Ada	Brier AUC	ECE	Ada	Brier AUC	ECE	Ada	Brier AUC	ECE	Ada	Brier AUC	ECE	Ada	Brier AUC
AI2D	0.271	0.269	0.288	0.746	0.133	0.131	0.219	0.746	0.133	0.140	0.219	0.747	0.059	0.082	0.201
ScienceQA	0.236	0.231	0.257	0.760	0.093	0.102	0.197	0.760	0.087	0.093	0.196	0.761	0.079	0.069	0.191
TextVQA	0.322	0.319	0.328	0.684	0.182	0.183	0.250	0.684	0.183	0.183	0.250	0.689	0.089	0.104	0.214
VizWiz-VQA	0.306	0.306	0.243	0.797	0.294	0.294	0.232	0.797	0.291	0.291	0.231	0.796	0.062	0.076	0.143
VQAv2	0.104	0.103	0.171	0.641	0.138	0.136	0.178	0.641	0.123	0.120	0.175	0.658	0.105	0.109	0.171
Macro Avg.	0.248	0.246	0.258	0.726	0.168	0.169	0.215	0.726	0.163	0.165	0.214	0.730	0.079	0.088	0.184

Table 7: **RQ1 Calibration Benchmark on MQT-LLaVA (7B)**. Comparison of calibration methods across 5 vision-language benchmarks. Metrics: Expected Calibration Error (ECE ↓), Adaptive ECE (Ada-ECE ↓), Brier Score (↓), and AUC (↑). **Bold**: Rank 1; *italic*: Rank 2 among single-pass methods.

1.5 RQ1 Master Table: Core 5 Vision-Language Calibration Benchmark (M3 & MQT 7B)

tabularx array

Calibration Method	Adversarial VQA				VLLM-Safety				Macro Average			
	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Panel A: M3-LLaVA (7B)												
Naive Confidence (NC)	0.2566	0.2540	0.3194	0.599	<i>0.1159</i>	<i>0.1050</i>	0.1732	0.814	0.1863	0.1795	0.2463	<i>0.707</i>
Temperature Scaling (TS)	<i>0.0445</i>	0.1796	0.2480	0.599	0.1698	0.1680	0.1812	0.814	0.1071	0.1738	0.2146	<i>0.707</i>
Adaptive TS (ATS)	0.0429	<i>0.1694</i>	0.2477	<i>0.601</i>	0.1408	0.1384	<i>0.1730</i>	<i>0.813</i>	<i>0.0918</i>	<i>0.1539</i>	<i>0.2104</i>	<i>0.707</i>
Platt Scaling (1D)	0.0684	0.1767	<i>0.2463</i>	0.599	0.1476	0.1404	0.1798	0.814	0.1080	0.1585	0.2131	<i>0.707</i>
LN-Entropy	-	-	-	-	-	-	-	-	-	-	-	-
Semantic Entropy	-	-	-	-	-	-	-	-	-	-	-	-
EigenScore	-	-	-	-	-	-	-	-	-	-	-	-
UMPIRE	-	-	-	-	-	-	-	-	-	-	-	-
Trajectory Platt (5D)	0.0526	0.0745	0.2240	0.691	0.0587	0.0688	0.1709	0.773	0.0557	0.0717	0.1975	0.732
Panel B: MQT-LLaVA (7B)												
Naive Confidence (NC)	0.2788	0.2800	0.3385	0.602	<i>0.0963</i>	<i>0.0974</i>	0.1966	<i>0.763</i>	0.1876	0.1887	0.2675	0.682
Temperature Scaling (TS)	0.0656	0.1328	0.2491	0.602	0.1420	0.1442	0.2054	<i>0.763</i>	0.1038	0.1385	0.2272	0.682
Adaptive TS (ATS)	0.0662	<i>0.1286</i>	0.2488	<i>0.604</i>	0.1213	0.1319	0.1966	0.768	0.0937	0.1303	0.2227	<i>0.686</i>
Platt Scaling (1D)	0.0118	0.1366	<i>0.2449</i>	0.602	0.1134	0.1129	<i>0.1937</i>	<i>0.763</i>	<i>0.0626</i>	<i>0.1248</i>	<i>0.2193</i>	0.682
LN-Entropy	-	-	-	-	-	-	-	-	-	-	-	-
Semantic Entropy	-	-	-	-	-	-	-	-	-	-	-	-
EigenScore	-	-	-	-	-	-	-	-	-	-	-	-
UMPIRE	-	-	-	-	-	-	-	-	-	-	-	-
Trajectory Platt (5D)	<i>0.0502</i>	0.0654	0.2332	0.654	0.0568	0.0658	0.1921	0.733	0.0535	0.0656	0.2127	0.694

Table 8: **RQ1 Calibration Robustness Under Adversarial and Safety Stress Benchmarks.** Performance on Adversarial VQA, VLLM-Safety, and Adversarial Macro Average for M3-LLaVA (7B) (Panel A) and MQT-LLaVA (7B) (Panel B). **Bold**: Rank 1; *italic*: Rank 2 within each metric column among single-pass methods.

1.6 RQ1 Master Table: Adversarial & Safety Benchmark (M3 & MQT 7B)

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0, K = 1$)	0.0864	0.0723	0.1387	0.772
Temperature Scaling (TS)	Single-Pass ($T = 0.0, K = 1$)	0.0940	0.0858	0.1417	0.772
Adaptive TS (ATS)	Single-Pass ($T = 0.0, K = 1$)	0.0839	0.0752	<i>0.1381</i>	0.788
Platt Scaling (1D)	Single-Pass ($T = 0.0, K = 1$)	0.0552	<i>0.0697</i>	0.1391	0.772
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	0.1240	-	-	0.787
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	0.1140	-	-	0.690
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	0.0620	-	-	<i>0.816</i>
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	<i>0.0580</i>	-	-	0.819
Trajectory Platt (5D)	Single-Pass ($T = 0.0, K = 1$)	0.0686	0.0650	0.1380	0.788

Table 9: **VQAv2 Calibration: Single-Pass Trajectory Calibration (1×) vs. Multi-Rollout UMPIRE Suite (10×) (M3-LLaVA 7B)**. Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0, K = 1$)	0.1049	<i>0.1031</i>	0.1713	0.641
Temperature Scaling (TS)	Single-Pass ($T = 0.0, K = 1$)	0.1387	0.1365	0.1789	0.641
Adaptive TS (ATS)	Single-Pass ($T = 0.0, K = 1$)	0.1236	0.1203	0.1753	0.658
Platt Scaling (1D)	Single-Pass ($T = 0.0, K = 1$)	0.1054	0.1094	<i>0.1712</i>	0.641
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	0.1230	-	-	0.767
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	0.1370	-	-	0.699
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	0.1020	-	-	<i>0.773</i>
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	<i>0.0780</i>	-	-	0.790
Trajectory Platt (5D)	Single-Pass ($T = 0.0, K = 1$)	0.0716	0.0764	0.1629	0.739

Table 10: **VQAv2 Calibration: Single-Pass Trajectory Calibration (1×) vs. Multi-Rollout UMPIRE Suite (10×) (MQT-LLaVA 7B)**. Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

1.7 Group 9: VQAv2 Single-Pass (1×) vs. Multi-Rollout UMPIRE Suite (10×)

Configuration / Source Anchor	M3-LLaVA (7B)				MQT-LLaVA (7B)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0477	0.0585	0.720	0.1356	0.0553	0.0642	0.743	0.1312
<i>Zero-Shot Transfer Baselines (Macro Mean Across All Pairs)</i>								
Naive Confidence (NC)	0.4157	0.4155	0.682	0.4076	0.3423	0.3418	<i>0.688</i>	0.3228
Temperature Scaling (TS)	0.3327	0.3478	0.682	<i>0.3102</i>	0.3010	0.3046	<i>0.688</i>	0.2581
Adaptive TS (ATS)	0.3328	0.3475	<i>0.686</i>	0.3094	0.3011	0.3046	0.689	0.2580
Platt Scaling (1D)	0.3688	0.3714	0.649	0.3531	0.3142	0.3151	0.594	0.2946
<i>Trajectory Platt (5D) Candidate Source Anchors</i>								
VQA_{v2}	0.4053	0.4055	0.687	0.3766	0.3381	0.3357	0.662	0.3020
VizWiz-VQA	0.4046	0.4048	0.671	0.3867	<i>0.2096</i>	<i>0.2088</i>	0.685	<i>0.2090</i>
DocVQA	<i>0.3003</i>	<i>0.2999</i>	0.683	0.3174	0.3788	0.3789	0.644	0.3686
TextVQA	0.4201	0.4199	0.676	0.4052	0.2068	0.2082	0.668	0.1868
ChartQA	0.2972	0.2989	0.672	0.3175	0.3609	0.3610	0.475	0.3557
POPE	0.4380	0.4382	0.677	0.4231	0.3429	0.3417	0.672	0.3221

Table 11: **Zero-Shot Cross-Domain Source Anchor Leaderboard.** Performance of Trajectory Platt (5D) across candidate source anchors vs. transfer baselines across 12 unseen target benchmarks on M3-LLaVA and MQT-LLaVA (7B). **Bold:** Rank 1; *Italic:* Rank 2 among transfer configurations. ↓/↑: lower/higher is better.

1.8 Group 6: Zero-Shot Source Anchor Leaderboard

1.9 Group 7: Sample Efficiency & Low-Data Calibration Scaling

Budget (N)	Calibration Method	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
Panel A: TextVQA (Scene Text VQA)					
$N = 50$	Naive Confidence (NC)	0.1050	0.1107	0.839	0.1714
	Temperature Scaling (TS)	0.1141	0.1143	0.839	0.1726
	Adaptive TS (ATS)	0.1189	0.1142	0.839	0.1725
	Platt Scaling (1D)	<i>0.0873</i>	<i>0.0800</i>	0.839	0.1621
	Trajectory Platt (5D)	0.0717	0.0743	<i>0.817</i>	<i>0.1711</i>
$N = 100$	Naive Confidence (NC)	0.1050	0.1107	0.839	0.1714
	Temperature Scaling (TS)	0.1089	0.1130	0.839	0.1719
	Adaptive TS (ATS)	0.1114	0.1093	0.839	0.1718
	Platt Scaling (1D)	<i>0.0875</i>	<i>0.0838</i>	0.839	0.1628
	Trajectory Platt (5D)	0.0764	0.0737	<i>0.829</i>	<i>0.1661</i>
$N = 200$	Naive Confidence (NC)	0.1050	0.1107	0.839	0.1714
	Temperature Scaling (TS)	0.1089	0.1137	0.839	0.1722
	Adaptive TS (ATS)	0.1128	0.1122	0.839	0.1720
	Platt Scaling (1D)	<i>0.0946</i>	<i>0.0908</i>	0.839	0.1648
	Trajectory Platt (5D)	0.0661	0.0700	<i>0.824</i>	<i>0.1691</i>
$N = 500$	Naive Confidence (NC)	0.1050	0.1107	0.839	0.1714
	Temperature Scaling (TS)	0.1068	0.1118	0.839	0.1714
	Adaptive TS (ATS)	0.1073	0.1115	0.839	0.1712
	Platt Scaling (1D)	<i>0.0835</i>	<i>0.0832</i>	0.839	0.1623
	Trajectory Platt (5D)	0.0652	0.0585	<i>0.832</i>	<i>0.1641</i>
$N = 1000$	Naive Confidence (NC)	0.1050	0.1107	0.839	0.1714
	Temperature Scaling (TS)	0.1012	0.1114	0.839	0.1712
	Adaptive TS (ATS)	0.1015	0.1141	0.839	0.1710
	Platt Scaling (1D)	<i>0.0849</i>	<i>0.0818</i>	0.839	0.1615
	Trajectory Platt (5D)	0.0624	0.0656	<i>0.833</i>	<i>0.1636</i>
$N = 1500$	Naive Confidence (NC)	0.1050	0.1107	0.839	0.1714
	Temperature Scaling (TS)	0.1016	0.1114	0.839	0.1712
	Adaptive TS (ATS)	0.0991	0.1131	0.839	0.1710
	Platt Scaling (1D)	<i>0.0853</i>	<i>0.0824</i>	0.839	0.1616
	Trajectory Platt (5D)	0.0563	0.0570	<i>0.834</i>	<i>0.1629</i>
Full	Naive Confidence (NC)	0.1050	0.1107	0.839	0.1714
	Temperature Scaling (TS)	0.1038	0.1114	0.839	0.1712
	Adaptive TS (ATS)	0.1015	0.1131	0.839	0.1710
	Platt Scaling (1D)	<i>0.0847</i>	<i>0.0822</i>	0.839	0.1615
	Trajectory Platt (5D)	0.0578	0.0535	<i>0.835</i>	<i>0.1628</i>
Panel B: VQAv2 (General Scene VQA)					
$N = 50$	Naive Confidence (NC)	0.0864	<i>0.0723</i>	<i>0.772</i>	0.1387
	Temperature Scaling (TS)	0.1052	0.0951	<i>0.772</i>	0.1435
	Adaptive TS (ATS)	0.0884	0.0829	0.788	<i>0.1394</i>
	Platt Scaling (1D)	0.0691	0.0790	<i>0.772</i>	0.1411
	Trajectory Platt (5D)	<i>0.0714</i>	0.0714	0.764	0.1444
$N = 100$	Naive Confidence (NC)	0.0864	<i>0.0723</i>	0.772	0.1387
	Temperature Scaling (TS)	0.1006	0.0910	0.772	0.1429
	Adaptive TS (ATS)	0.0879	0.0821	0.782	<i>0.1401</i>
	Platt Scaling (1D)	<i>0.0723</i>	0.0776	0.772	0.1408
	Trajectory Platt (5D)	0.0595	0.0681	<i>0.775</i>	<i>0.1401</i>
$N = 200$	Naive Confidence (NC)	0.0864	<i>0.0723</i>	0.772	0.1387
	Temperature Scaling (TS)	0.0959	0.0871	0.772	0.1419
	Adaptive TS (ATS)	0.0786	0.0801	0.786	<i>0.1389</i>
	Platt Scaling (1D)	0.0590	0.0714	0.772	0.1392
	Trajectory Platt (5D)	<i>0.0665</i>	0.0733	<i>0.785</i>	0.1391
$N = 500$	Naive Confidence (NC)	0.0864	0.0723	0.772	0.1387
	Temperature Scaling (TS)	0.0908	0.0826	0.772	0.1410
	Adaptive TS (ATS)	0.0830	0.0770	<i>0.785</i>	<i>0.1383</i>
	Platt Scaling (1D)	0.0536	<i>0.0670</i>	0.772	0.1381
	Trajectory Platt (5D)	<i>0.0620</i>	0.0639	0.787	0.1381
$N = 1000$	Naive Confidence (NC)	0.0864	0.0723	0.772	0.1387
	Temperature Scaling (TS)	0.0932	0.0852	0.772	0.1416
	Adaptive TS (ATS)	0.0806	0.0740	0.788	0.1382
	Platt Scaling (1D)	0.0575	<i>0.0693</i>	0.772	0.1390
	Trajectory Platt (5D)	<i>0.0668</i>	0.0690	<i>0.787</i>	<i>0.1386</i>
$N = 1500$	Naive Confidence (NC)	0.0864	0.0723	<i>0.772</i>	<i>0.1387</i>
	Temperature Scaling (TS)	0.0939	0.0856	<i>0.772</i>	0.1417
	Adaptive TS (ATS)	0.0814	0.0716	0.788	0.1381
	Platt Scaling (1D)	0.0575	<i>0.0695</i>	<i>0.772</i>	0.1391
	Trajectory Platt (5D)	<i>0.0664</i>	0.0668	0.788	0.1381
Full	Naive Confidence (NC)	0.0864	0.0723	<i>0.772</i>	0.1387
	Temperature Scaling (TS)	0.0940	0.0858	<i>0.772</i>	0.1417
	Adaptive TS (ATS)	0.0839	0.0752	0.788	<i>0.1381</i>
	Platt Scaling (1D)	0.0552	<i>0.0697</i>	<i>0.772</i>	0.1391
	Trajectory Platt (5D)	<i>0.0686</i>	0.0650	0.788	0.1380

Table 12: **Calibration Sample Efficiency Scaling Across Training Budgets (M3-LLaVA 7B).** Evaluated across 7 training budgets $N \in \{50, 100, 200, 500, 1000, 1500, \text{Full}\}$ over 5 subsampling seeds. Panel A: TextVQA; Panel B: VQAv2. **Bold:** best; *italic:* second best within each budget tier.

Budget (N)	Calibration Method	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
Panel A: TextVQA (Scene Text VQA)					
$N = 50$	Naive Confidence (NC)	0.3227	0.3191	0.684	0.3286
	Temperature Scaling (TS)	0.1993	0.1996	0.684	0.2539
	Adaptive TS (ATS)	0.2059	0.2077	<i>0.689</i>	0.2549
	Platt Scaling (1D)	<i>0.1184</i>	<i>0.1207</i>	0.684	<i>0.2232</i>
	Trajectory Platt (5D)	0.1026	0.1071	0.694	0.2097
$N = 100$	Naive Confidence (NC)	0.3227	0.3191	0.684	0.3286
	Temperature Scaling (TS)	0.1963	0.1968	0.684	0.2534
	Adaptive TS (ATS)	0.1985	0.1985	<i>0.687</i>	0.2535
	Platt Scaling (1D)	<i>0.1034</i>	<i>0.1082</i>	0.684	<i>0.2184</i>
	Trajectory Platt (5D)	0.0696	0.0790	0.705	0.2018
$N = 200$	Naive Confidence (NC)	0.3227	0.3191	0.684	0.3286
	Temperature Scaling (TS)	0.1860	0.1867	0.684	0.2509
	Adaptive TS (ATS)	0.1873	0.1873	<i>0.688</i>	0.2507
	Platt Scaling (1D)	<i>0.1001</i>	<i>0.1058</i>	0.684	<i>0.2162</i>
	Trajectory Platt (5D)	0.0684	0.0721	0.711	0.1998
$N = 500$	Naive Confidence (NC)	0.3227	0.3191	0.684	0.3286
	Temperature Scaling (TS)	0.1894	0.1895	0.684	0.2514
	Adaptive TS (ATS)	0.1914	0.1914	<i>0.690</i>	0.2511
	Platt Scaling (1D)	<i>0.1058</i>	<i>0.1078</i>	0.684	<i>0.2170</i>
	Trajectory Platt (5D)	0.0661	0.0846	0.717	0.1962
$N = 1000$	Naive Confidence (NC)	0.3227	0.3191	0.684	0.3286
	Temperature Scaling (TS)	0.1859	0.1866	0.684	0.2509
	Adaptive TS (ATS)	0.1875	0.1875	<i>0.689</i>	0.2507
	Platt Scaling (1D)	<i>0.0975</i>	<i>0.1058</i>	0.684	<i>0.2158</i>
	Trajectory Platt (5D)	0.0590	0.0697	0.718	0.1956
$N = 1500$	Naive Confidence (NC)	0.3227	0.3191	0.684	0.3286
	Temperature Scaling (TS)	0.1825	0.1835	0.684	0.2503
	Adaptive TS (ATS)	0.1840	0.1840	<i>0.689</i>	0.2501
	Platt Scaling (1D)	<i>0.0881</i>	<i>0.1045</i>	0.684	<i>0.2148</i>
	Trajectory Platt (5D)	0.0551	0.0739	0.719	0.1947
Full	Naive Confidence (NC)	0.3227	0.3191	0.684	0.3286
	Temperature Scaling (TS)	0.1821	0.1832	0.684	0.2503
	Adaptive TS (ATS)	0.1836	0.1836	<i>0.689</i>	0.2500
	Platt Scaling (1D)	<i>0.0895</i>	<i>0.1046</i>	0.684	<i>0.2148</i>
	Trajectory Platt (5D)	0.0525	0.0776	0.719	0.1948
Panel B: VQAv2 (General Scene VQA)					
$N = 50$	Naive Confidence (NC)	<i>0.1049</i>	<i>0.1031</i>	0.641	0.1713
	Temperature Scaling (TS)	0.1425	0.1394	0.641	0.1796
	Adaptive TS (ATS)	0.1213	0.1133	<i>0.666</i>	0.1735
	Platt Scaling (1D)	0.1076	0.1161	0.641	0.1723
	Trajectory Platt (5D)	0.0840	0.0939	0.706	<i>0.1722</i>
$N = 100$	Naive Confidence (NC)	0.1049	<i>0.1031</i>	0.641	0.1713
	Temperature Scaling (TS)	0.1386	0.1367	0.641	0.1785
	Adaptive TS (ATS)	0.1177	0.1147	<i>0.665</i>	0.1736
	Platt Scaling (1D)	<i>0.1022</i>	0.1106	0.641	<i>0.1712</i>
	Trajectory Platt (5D)	0.0827	0.0849	0.730	0.1683
$N = 200$	Naive Confidence (NC)	<i>0.1049</i>	<i>0.1031</i>	0.641	<i>0.1713</i>
	Temperature Scaling (TS)	0.1429	0.1403	0.641	0.1796
	Adaptive TS (ATS)	0.1176	0.1159	<i>0.663</i>	0.1746
	Platt Scaling (1D)	0.1096	0.1134	0.641	0.1716
	Trajectory Platt (5D)	0.0679	0.0800	0.739	0.1647
$N = 500$	Naive Confidence (NC)	<i>0.1049</i>	<i>0.1031</i>	0.641	<i>0.1713</i>
	Temperature Scaling (TS)	0.1508	0.1470	0.641	0.1812
	Adaptive TS (ATS)	0.1322	0.1278	<i>0.659</i>	0.1765
	Platt Scaling (1D)	0.1086	0.1179	0.641	0.1721
	Trajectory Platt (5D)	0.0643	0.0743	0.748	0.1616
$N = 1000$	Naive Confidence (NC)	<i>0.1049</i>	<i>0.1031</i>	0.641	<i>0.1713</i>
	Temperature Scaling (TS)	0.1434	0.1392	0.641	0.1796
	Adaptive TS (ATS)	0.1248	0.1223	<i>0.658</i>	0.1758
	Platt Scaling (1D)	0.1073	0.1122	0.641	0.1715
	Trajectory Platt (5D)	0.0609	0.0709	0.741	0.1625
$N = 1500$	Naive Confidence (NC)	<i>0.1049</i>	<i>0.1031</i>	0.641	<i>0.1713</i>
	Temperature Scaling (TS)	0.1401	0.1373	0.641	0.1791
	Adaptive TS (ATS)	0.1218	0.1200	<i>0.658</i>	0.1754
	Platt Scaling (1D)	0.1096	0.1102	0.641	<i>0.1713</i>
	Trajectory Platt (5D)	0.0691	0.0778	0.740	0.1627
Full	Naive Confidence (NC)	<i>0.1049</i>	<i>0.1031</i>	0.641	0.1713
	Temperature Scaling (TS)	0.1387	0.1365	0.641	0.1789
	Adaptive TS (ATS)	0.1236	0.1203	<i>0.658</i>	0.1753
	Platt Scaling (1D)	0.1054	0.1094	0.641	<i>0.1712</i>
	Trajectory Platt (5D)	0.0716	0.0764	0.739	0.1629

Table 13: **Calibration Sample Efficiency Scaling Across Training Budgets (MQT-LLaVA 7B).** Evaluated across 7 training budgets $N \in \{50, 100, 200, 500, 1000, 1500, \text{Full}\}$ over 5 subsampling seeds. Panel A: TextVQA; Panel B: VQAv2. **Bold:** best; *italic:* second best within each budget tier.

Calibration Method	Transfer Protocol	Macro ECE ↓	Macro Ada-ECE ↓	Macro Brier ↓	Macro AUROC
Naive Confidence (NC)	Zero-Shot Base	0.1940	0.1939	0.2356	<i>0.765</i>
Naive Confidence (NC)	Target Adapted (Saerens-EM)	0.3757	0.3757	0.3669	0.713
Temperature Scaling (TS)	Zero-Shot Base	0.1541	0.2132	0.2363	<i>0.765</i>
Temperature Scaling (TS)	Target Adapted (Saerens-EM)	0.4061	0.4061	0.4061	<i>0.765</i>
Adaptive TS (ATS)	Zero-Shot Base	<i>0.1544</i>	0.2172	<i>0.2362</i>	0.767
Adaptive TS (ATS)	Target Adapted (Saerens-EM)	0.4061	0.4061	0.4061	0.767
Platt Scaling (1D)	Zero-Shot Base	0.2443	0.2541	0.2800	<i>0.765</i>
Platt Scaling (1D)	Target Adapted (Saerens-EM)	0.5623	0.5623	0.5623	<i>0.765</i>
Trajectory Platt (5D)	Zero-Shot Base	0.1983	<i>0.2003</i>	0.2467	0.723
Trajectory Platt (5D)	Target Adapted (Saerens-EM)	0.4167	0.4180	0.4131	0.723

Table 14: **Leave-One-Dataset-Out (LODO) Cross-Domain Transfer (M3-LLaVA 7B)**. Macro-averaged across held-out Core 5 benchmarks. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Transfer Protocol	Macro ECE ↓	Macro Ada-ECE ↓	Macro Brier ↓	Macro AUROC
Naive Confidence (NC)	Zero-Shot Base	0.2448	0.2448	0.2542	<i>0.753</i>
Naive Confidence (NC)	Target Adapted (Saerens-EM)	0.4664	0.4664	0.4495	0.673
Temperature Scaling (TS)	Zero-Shot Base	0.1950	0.1983	0.2247	<i>0.753</i>
Temperature Scaling (TS)	Target Adapted (Saerens-EM)	0.4956	0.4956	0.4956	<i>0.753</i>
Adaptive TS (ATS)	Zero-Shot Base	0.1926	0.1962	0.2233	0.756
Adaptive TS (ATS)	Target Adapted (Saerens-EM)	0.4956	0.4956	0.4956	0.756
Platt Scaling (1D)	Zero-Shot Base	<i>0.1480</i>	<i>0.1482</i>	<i>0.2114</i>	<i>0.753</i>
Platt Scaling (1D)	Target Adapted (Saerens-EM)	0.2414	0.2408	0.2617	<i>0.753</i>
Trajectory Platt (5D)	Zero-Shot Base	0.1233	0.1226	0.2070	0.734
Trajectory Platt (5D)	Target Adapted (Saerens-EM)	0.2152	0.2157	0.2567	0.734

Table 15: **Leave-One-Dataset-Out (LODO) Cross-Domain Transfer (MQT-LLaVA 7B)**. Macro-averaged across held-out Core 5 benchmarks. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

1.10 Group 4: Leave-One-Dataset-Out (LODO) Cross-Domain Transfer

2 Group 3: Temperature Transfer Robustness Ablation

Evaluating calibrators trained at $T = 0.0$ and tested across $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and temperature Mean.

2.1 Temperature Transfer: Macro-Averaged Across 4 Benchmarks

Calibration Method	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	0.1302	0.1334	0.1759	0.794
Temperature Scaling (TS)	0.1071	0.1082	0.1646	0.794
Adaptive TS (ATS)	0.1061	0.1086	<i>0.1641</i>	<i>0.797</i>
Platt Scaling (1D)	<i>0.0602</i>	0.0583	0.1515	0.794
Trajectory Platt (5D)	0.0554	<i>0.0617</i>	0.1515	0.798
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	0.2846	0.2771	0.2767	0.780
Temperature Scaling (TS)	0.2561	0.2492	0.2486	0.780
Adaptive TS (ATS)	0.2550	0.2492	0.2486	<i>0.782</i>
Platt Scaling (1D)	<i>0.2360</i>	<i>0.2285</i>	<i>0.2359</i>	0.780
Trajectory Platt (5D)	0.2247	0.2183	0.2308	0.784
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	0.1897	0.1802	0.2045	0.799
Temperature Scaling (TS)	0.1484	0.1449	0.1815	0.799
Adaptive TS (ATS)	0.1477	0.1480	0.1811	<i>0.802</i>
Platt Scaling (1D)	<i>0.1145</i>	<i>0.1147</i>	<i>0.1693</i>	0.799
Trajectory Platt (5D)	0.1107	0.1051	0.1668	0.803
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	0.1189	0.1175	0.1582	<i>0.840</i>
Temperature Scaling (TS)	0.0954	0.0941	0.1458	<i>0.840</i>
Adaptive TS (ATS)	0.0978	0.0971	0.1452	0.841
Platt Scaling (1D)	0.0578	0.0520	0.1351	<i>0.840</i>
Trajectory Platt (5D)	<i>0.0767</i>	<i>0.0701</i>	<i>0.1371</i>	<i>0.840</i>
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	0.0992	0.1003	0.1426	<i>0.841</i>
Temperature Scaling (TS)	<i>0.0925</i>	0.0907	<i>0.1390</i>	<i>0.841</i>
Adaptive TS (ATS)	0.0930	0.0941	0.1386	0.843
Platt Scaling (1D)	0.0907	<i>0.0912</i>	<i>0.1390</i>	<i>0.841</i>
Trajectory Platt (5D)	0.1092	0.1068	0.1464	0.840
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	0.1645	0.1617	0.1916	<i>0.811</i>
Temperature Scaling (TS)	0.1399	0.1374	0.1759	<i>0.811</i>
Adaptive TS (ATS)	0.1399	0.1394	0.1755	0.813
Platt Scaling (1D)	0.1119	0.1089	0.1662	<i>0.811</i>
Trajectory Platt (5D)	<i>0.1153</i>	<i>0.1124</i>	<i>0.1665</i>	0.813

Table 16: **Temperature Transfer Robustness (Macro-Averaged Across 4 Benchmarks) (M3-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	0.2265	0.2251	0.2301	0.779
Temperature Scaling (TS)	0.1511	0.1572	0.1927	0.779
Adaptive TS (ATS)	0.1499	0.1544	0.1920	<i>0.780</i>
Platt Scaling (1D)	<i>0.0679</i>	<i>0.0741</i>	<i>0.1598</i>	0.779
Trajectory Platt (5D)	0.0532	0.0653	0.1532	0.789
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	0.4112	0.4092	0.3853	0.758
Temperature Scaling (TS)	0.2826	0.2820	0.2669	0.758
Adaptive TS (ATS)	0.2873	0.2875	0.2682	<i>0.762</i>
Platt Scaling (1D)	<i>0.2489</i>	<i>0.2442</i>	<i>0.2500</i>	0.758
Trajectory Platt (5D)	0.2066	0.2061	0.2198	0.786
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	0.3452	0.3417	0.3170	0.763
Temperature Scaling (TS)	0.2179	0.2189	0.2245	0.763
Adaptive TS (ATS)	0.2186	0.2195	0.2250	<i>0.765</i>
Platt Scaling (1D)	<i>0.1350</i>	<i>0.1328</i>	<i>0.1855</i>	0.763
Trajectory Platt (5D)	0.1048	0.1051	0.1633	0.794
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	0.2572	0.2583	0.2443	0.789
Temperature Scaling (TS)	0.1781	0.1831	0.1988	0.789
Adaptive TS (ATS)	0.1749	0.1821	0.1979	<i>0.791</i>
Platt Scaling (1D)	<i>0.0714</i>	<i>0.0708</i>	<i>0.1529</i>	0.789
Trajectory Platt (5D)	0.0503	0.0513	0.1393	0.817
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	0.2045	0.2069	0.2017	0.805
Temperature Scaling (TS)	0.1986	0.1993	0.1983	0.805
Adaptive TS (ATS)	0.1952	0.1988	0.1962	<i>0.806</i>
Platt Scaling (1D)	<i>0.0872</i>	0.0861	<i>0.1507</i>	0.805
Trajectory Platt (5D)	0.0779	<i>0.0869</i>	0.1439	0.820
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	0.2889	0.2882	0.2757	0.779
Temperature Scaling (TS)	0.2057	0.2081	0.2162	0.779
Adaptive TS (ATS)	0.2052	0.2085	0.2159	<i>0.781</i>
Platt Scaling (1D)	<i>0.1221</i>	<i>0.1216</i>	<i>0.1798</i>	0.779
Trajectory Platt (5D)	0.0986	0.1029	0.1639	0.801

Table 17: **Temperature Transfer Robustness (Macro-Averaged Across 4 Benchmarks) (MQT-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

2.2 Temperature Transfer: POPE

Calibration Method	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	0.0345	0.0440	<i>0.0882</i>	0.739
Temperature Scaling (TS)	0.0427	0.0437	0.0883	0.739
Adaptive TS (ATS)	0.0442	0.0427	0.0879	<i>0.750</i>
Platt Scaling (1D)	<i>0.0419</i>	<i>0.0432</i>	<i>0.0882</i>	0.739
Trajectory Platt (5D)	0.0482	0.0662	0.0887	0.751
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	0.0983	0.0869	0.0980	0.764
Temperature Scaling (TS)	<i>0.0969</i>	<i>0.0853</i>	<i>0.0972</i>	0.764
Adaptive TS (ATS)	0.0966	0.0854	0.0968	0.772
Platt Scaling (1D)	<i>0.0969</i>	0.0855	<i>0.0972</i>	0.764
Trajectory Platt (5D)	0.0971	0.0843	0.0975	<i>0.770</i>
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	0.0700	0.0653	0.0912	0.786
Temperature Scaling (TS)	0.0673	0.0602	0.0904	0.786
Adaptive TS (ATS)	0.0640	0.0642	0.0901	0.796
Platt Scaling (1D)	<i>0.0664</i>	<i>0.0597</i>	0.0904	0.786
Trajectory Platt (5D)	0.0745	0.0576	<i>0.0902</i>	<i>0.794</i>
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	0.0434	0.0456	<i>0.0882</i>	0.826
Temperature Scaling (TS)	<i>0.0508</i>	<i>0.0443</i>	0.0883	0.826
Adaptive TS (ATS)	0.0515	0.0438	<i>0.0882</i>	<i>0.828</i>
Platt Scaling (1D)	0.0516	0.0447	0.0883	0.826
Trajectory Platt (5D)	0.0677	0.0482	0.0872	0.832
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	0.0538	0.0614	0.0953	0.876
Temperature Scaling (TS)	<i>0.0683</i>	<i>0.0680</i>	<i>0.0970</i>	0.876
Adaptive TS (ATS)	0.0734	0.0798	0.0977	0.880
Platt Scaling (1D)	0.0725	0.0681	0.0972	0.876
Trajectory Platt (5D)	0.0971	0.0939	0.1004	<i>0.879</i>
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	0.0600	0.0606	<i>0.0922</i>	<i>0.798</i>
Temperature Scaling (TS)	<i>0.0652</i>	<i>0.0603</i>	<i>0.0922</i>	<i>0.798</i>
Adaptive TS (ATS)	0.0659	0.0632	0.0921	0.805
Platt Scaling (1D)	0.0659	0.0602	0.0923	<i>0.798</i>
Trajectory Platt (5D)	0.0769	0.0700	0.0928	0.805

Table 18: **Temperature Transfer Robustness on pope (M3-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best.

↓/↑: lower/higher is better.

Calibration Method	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	0.0401	0.0430	0.0903	0.876
Temperature Scaling (TS)	0.0352	0.0493	<i>0.0902</i>	0.876
Adaptive TS (ATS)	0.0375	0.0490	<i>0.0902</i>	<i>0.874</i>
Platt Scaling (1D)	0.0397	0.0464	0.0893	0.876
Trajectory Platt (5D)	<i>0.0355</i>	<i>0.0448</i>	0.0903	0.873
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	0.1043	0.1043	0.1108	0.872
Temperature Scaling (TS)	<i>0.0942</i>	<i>0.0918</i>	<i>0.1043</i>	0.872
Adaptive TS (ATS)	0.0947	0.0926	0.1048	<i>0.871</i>
Platt Scaling (1D)	0.0920	0.0891	0.1031	0.872
Trajectory Platt (5D)	0.1026	0.1017	0.1108	0.868
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	0.0853	0.0826	0.1098	0.852
Temperature Scaling (TS)	<i>0.0612</i>	0.0616	<i>0.1036</i>	0.852
Adaptive TS (ATS)	0.0610	<i>0.0615</i>	0.1038	<i>0.851</i>
Platt Scaling (1D)	0.0678	0.0566	0.1026	0.852
Trajectory Platt (5D)	0.0767	0.0688	0.1068	0.846
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	0.0539	0.0584	0.1081	0.855
Temperature Scaling (TS)	0.0431	0.0494	<i>0.1046</i>	0.855
Adaptive TS (ATS)	<i>0.0397</i>	0.0559	<i>0.1046</i>	0.855
Platt Scaling (1D)	0.0442	<i>0.0483</i>	0.1037	0.855
Trajectory Platt (5D)	0.0234	0.0368	0.1048	<i>0.852</i>
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	0.0846	0.0936	0.1545	0.810
Temperature Scaling (TS)	0.0873	0.0736	0.1497	0.810
Adaptive TS (ATS)	0.0847	0.0705	0.1496	0.810
Platt Scaling (1D)	<i>0.0648</i>	<i>0.0712</i>	0.1473	0.810
Trajectory Platt (5D)	0.0620	0.0743	<i>0.1487</i>	<i>0.805</i>
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	0.0736	0.0764	0.1147	0.853
Temperature Scaling (TS)	0.0642	<i>0.0652</i>	<i>0.1105</i>	0.853
Adaptive TS (ATS)	0.0635	0.0659	0.1106	<i>0.852</i>
Platt Scaling (1D)	<i>0.0617</i>	0.0623	0.1092	0.853
Trajectory Platt (5D)	0.0601	0.0653	0.1123	0.849

Table 19: **Temperature Transfer Robustness** on pope (MQT-LLaVA 7B). Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

2.3 Temperature Transfer: ScienceQA

Calibration Method	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	0.2216	0.2193	0.2368	0.807
Temperature Scaling (TS)	0.1310	0.1268	0.1924	0.807
Adaptive TS (ATS)	0.1298	0.1300	0.1918	<i>0.808</i>
Platt Scaling (1D)	0.0480	0.0499	<i>0.1732</i>	0.807
Trajectory Platt (5D)	<i>0.0517</i>	<i>0.0615</i>	0.1719	0.813
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	0.3297	0.3297	0.3341	0.786
Temperature Scaling (TS)	<i>0.2555</i>	0.2561	<i>0.2658</i>	0.786
Adaptive TS (ATS)	0.2541	0.2548	0.2640	<i>0.792</i>
Platt Scaling (1D)	0.2558	<i>0.2508</i>	0.2660	0.786
Trajectory Platt (5D)	0.2562	0.2485	<i>0.2658</i>	0.793
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	0.2933	0.2902	0.2947	<i>0.797</i>
Temperature Scaling (TS)	0.1872	0.1870	0.2211	<i>0.797</i>
Adaptive TS (ATS)	0.1860	0.1893	0.2199	0.799
Platt Scaling (1D)	<i>0.1617</i>	<i>0.1677</i>	0.2082	<i>0.797</i>
Trajectory Platt (5D)	0.1602	0.1624	<i>0.2093</i>	0.799
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	0.2479	0.2447	0.2492	<i>0.817</i>
Temperature Scaling (TS)	0.1507	0.1482	0.1985	<i>0.817</i>
Adaptive TS (ATS)	0.1481	0.1588	0.1971	0.820
Platt Scaling (1D)	0.0605	0.0567	0.1726	<i>0.817</i>
Trajectory Platt (5D)	<i>0.0814</i>	<i>0.0744</i>	<i>0.1757</i>	0.811
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	0.1884	0.1907	0.2272	<i>0.789</i>
Temperature Scaling (TS)	0.1383	0.1384	0.2068	<i>0.789</i>
Adaptive TS (ATS)	0.1418	0.1476	0.2052	0.795
Platt Scaling (1D)	<i>0.0920</i>	<i>0.0890</i>	0.1905	<i>0.789</i>
Trajectory Platt (5D)	0.0854	0.0791	<i>0.1919</i>	0.783
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	0.2562	0.2549	0.2684	0.799
Temperature Scaling (TS)	0.1725	0.1713	0.2169	0.799
Adaptive TS (ATS)	0.1720	0.1761	0.2156	0.803
Platt Scaling (1D)	0.1236	0.1228	0.2021	0.799
Trajectory Platt (5D)	<i>0.1270</i>	<i>0.1252</i>	<i>0.2029</i>	<i>0.800</i>

Table 20: **Temperature Transfer Robustness on scienceqa (M3-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	0.2365	0.2317	0.2579	0.760
Temperature Scaling (TS)	0.0932	0.1023	0.1975	0.760
Adaptive TS (ATS)	0.0873	0.0938	0.1966	<i>0.761</i>
Platt Scaling (1D)	<i>0.0799</i>	<i>0.0691</i>	<i>0.1916</i>	0.760
Trajectory Platt (5D)	0.0646	0.0572	0.1860	0.769
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	0.3472	0.3393	0.3405	<i>0.740</i>
Temperature Scaling (TS)	0.2317	0.2317	0.2567	<i>0.740</i>
Adaptive TS (ATS)	0.2278	<i>0.2278</i>	0.2540	0.731
Platt Scaling (1D)	0.2339	0.2236	<i>0.2554</i>	<i>0.740</i>
Trajectory Platt (5D)	<i>0.2304</i>	0.2295	0.2575	0.756
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	0.3038	0.2926	0.3037	0.757
Temperature Scaling (TS)	0.1469	0.1504	0.2163	0.757
Adaptive TS (ATS)	0.1441	0.1473	0.2141	<i>0.759</i>
Platt Scaling (1D)	<i>0.1317</i>	0.1245	<i>0.2105</i>	0.757
Trajectory Platt (5D)	0.1166	<i>0.1250</i>	0.2056	0.768
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	0.2442	0.2442	0.2651	0.767
Temperature Scaling (TS)	0.1057	0.1196	0.2000	0.767
Adaptive TS (ATS)	0.0972	0.1041	0.1978	<i>0.770</i>
Platt Scaling (1D)	<i>0.0707</i>	<i>0.0682</i>	<i>0.1902</i>	0.767
Trajectory Platt (5D)	0.0628	0.0531	0.1838	0.777
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	0.1763	0.1771	0.2298	0.767
Temperature Scaling (TS)	<i>0.1088</i>	0.1264	<i>0.2020</i>	0.767
Adaptive TS (ATS)	0.1066	0.1318	0.2005	<i>0.770</i>
Platt Scaling (1D)	0.1293	0.1297	0.2062	0.767
Trajectory Platt (5D)	0.1283	<i>0.1286</i>	0.2033	0.775
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	0.2616	0.2570	0.2794	<i>0.758</i>
Temperature Scaling (TS)	0.1373	0.1461	0.2145	<i>0.758</i>
Adaptive TS (ATS)	0.1326	0.1409	0.2126	<i>0.758</i>
Platt Scaling (1D)	<i>0.1291</i>	<i>0.1230</i>	<i>0.2108</i>	<i>0.758</i>
Trajectory Platt (5D)	0.1205	0.1187	0.2072	0.769

Table 21: **Temperature Transfer Robustness on scienceqa (MQT-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

2.4 Temperature Transfer: TextVQA

Calibration Method	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	0.1050	0.1107	0.1714	0.839
Temperature Scaling (TS)	0.1038	0.1114	0.1712	0.839
Adaptive TS (ATS)	0.1015	0.1131	0.1710	0.839
Platt Scaling (1D)	<i>0.0847</i>	<i>0.0822</i>	0.1615	0.839
Trajectory Platt (5D)	0.0578	0.0535	<i>0.1628</i>	<i>0.835</i>
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	0.2703	0.2561	0.2691	0.804
Temperature Scaling (TS)	0.2609	0.2482	0.2611	0.804
Adaptive TS (ATS)	0.2561	0.2461	0.2584	<i>0.805</i>
Platt Scaling (1D)	<i>0.2452</i>	<i>0.2297</i>	<i>0.2510</i>	0.804
Trajectory Platt (5D)	0.2029	0.2014	0.2314	0.808
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	0.1585	0.1407	0.1849	<i>0.846</i>
Temperature Scaling (TS)	0.1376	0.1327	0.1815	<i>0.846</i>
Adaptive TS (ATS)	0.1412	0.1321	0.1802	<i>0.846</i>
Platt Scaling (1D)	<i>0.1119</i>	<i>0.1087</i>	<i>0.1697</i>	<i>0.846</i>
Trajectory Platt (5D)	0.0748	0.0667	0.1575	0.852
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	<i>0.0685</i>	<i>0.0687</i>	0.1256	<i>0.902</i>
Temperature Scaling (TS)	0.0705	0.0779	0.1263	<i>0.902</i>
Adaptive TS (ATS)	0.0855	0.0843	0.1272	0.901
Platt Scaling (1D)	0.0451	0.0461	0.1207	<i>0.902</i>
Trajectory Platt (5D)	0.0796	0.0761	<i>0.1254</i>	0.904
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	<i>0.0851</i>	0.0863	0.1221	0.871
Temperature Scaling (TS)	0.0826	0.0891	<i>0.1233</i>	0.871
Adaptive TS (ATS)	0.0876	<i>0.0873</i>	0.1244	0.871
Platt Scaling (1D)	0.1044	0.1132	0.1313	0.871
Trajectory Platt (5D)	0.1583	0.1577	0.1555	0.871
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	0.1375	0.1325	0.1746	<i>0.852</i>
Temperature Scaling (TS)	0.1311	0.1319	0.1727	<i>0.852</i>
Adaptive TS (ATS)	0.1344	0.1326	0.1722	<i>0.852</i>
Platt Scaling (1D)	<i>0.1183</i>	<i>0.1160</i>	<i>0.1669</i>	<i>0.852</i>
Trajectory Platt (5D)	0.1147	0.1111	0.1665	0.854

Table 22: **Temperature Transfer Robustness** on textvqa (M3-LLaVA 7B). Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	0.3227	0.3191	0.3286	0.684
Temperature Scaling (TS)	0.1821	0.1832	0.2503	0.684
Adaptive TS (ATS)	0.1836	0.1836	0.2500	<i>0.689</i>
Platt Scaling (1D)	<i>0.0895</i>	<i>0.1046</i>	<i>0.2148</i>	0.684
Trajectory Platt (5D)	0.0525	0.0776	0.1948	0.719
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	0.5645	0.5645	0.5294	0.663
Temperature Scaling (TS)	<i>0.3004</i>	0.3004	0.3007	0.663
Adaptive TS (ATS)	0.3155	0.3185	<i>0.2997</i>	<i>0.698</i>
Platt Scaling (1D)	0.3012	<i>0.2953</i>	0.3187	0.663
Trajectory Platt (5D)	0.1654	0.1663	0.2147	0.761
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	0.4808	0.4808	0.4405	0.674
Temperature Scaling (TS)	0.2532	0.2532	0.2724	0.674
Adaptive TS (ATS)	0.2549	0.2549	0.2714	<i>0.683</i>
Platt Scaling (1D)	<i>0.1680</i>	<i>0.1776</i>	<i>0.2479</i>	0.674
Trajectory Platt (5D)	0.0906	0.0816	0.1763	0.774
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	0.3738	0.3738	0.3509	0.698
Temperature Scaling (TS)	0.2265	0.2265	0.2534	0.698
Adaptive TS (ATS)	0.2273	0.2330	0.2527	<i>0.706</i>
Platt Scaling (1D)	<i>0.1082</i>	<i>0.1061</i>	<i>0.1997</i>	0.698
Trajectory Platt (5D)	0.0478	0.0547	0.1538	0.802
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	0.3124	0.3124	0.2717	0.759
Temperature Scaling (TS)	0.2801	0.2801	0.2442	0.759
Adaptive TS (ATS)	0.2806	0.2848	0.2442	<i>0.762</i>
Platt Scaling (1D)	<i>0.0841</i>	<i>0.0792</i>	<i>0.1578</i>	0.759
Trajectory Platt (5D)	0.0566	0.0685	0.1333	0.810
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	0.4108	0.4101	0.3842	0.696
Temperature Scaling (TS)	0.2484	0.2487	0.2642	0.696
Adaptive TS (ATS)	0.2524	0.2550	0.2636	<i>0.708</i>
Platt Scaling (1D)	<i>0.1502</i>	<i>0.1526</i>	<i>0.2278</i>	0.696
Trajectory Platt (5D)	0.0826	0.0897	0.1746	0.773

Table 23: **Temperature Transfer Robustness on textvqa (MQT-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

2.5 Temperature Transfer: VizWiz-VQA

Calibration Method	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	0.1597	0.1597	0.2071	<i>0.793</i>
Temperature Scaling (TS)	0.1509	0.1509	0.2065	<i>0.793</i>
Adaptive TS (ATS)	0.1487	0.1487	<i>0.2058</i>	<i>0.793</i>
Platt Scaling (1D)	<i>0.0664</i>	0.0580	0.1828	<i>0.793</i>
Trajectory Platt (5D)	0.0638	<i>0.0657</i>	0.1828	0.794
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	0.4403	0.4359	0.4055	0.764
Temperature Scaling (TS)	0.4110	0.4070	0.3701	0.764
Adaptive TS (ATS)	0.4133	0.4105	0.3751	0.762
Platt Scaling (1D)	<i>0.3462</i>	<i>0.3481</i>	<i>0.3295</i>	0.764
Trajectory Platt (5D)	0.3427	0.3391	0.3284	<i>0.763</i>
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	0.2372	0.2245	0.2470	0.768
Temperature Scaling (TS)	0.2017	0.1998	0.2331	0.768
Adaptive TS (ATS)	0.1998	0.2062	0.2340	0.768
Platt Scaling (1D)	0.1180	0.1225	0.2090	0.768
Trajectory Platt (5D)	<i>0.1333</i>	<i>0.1337</i>	<i>0.2103</i>	<i>0.766</i>
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	0.1158	0.1110	0.1699	<i>0.815</i>
Temperature Scaling (TS)	0.1095	0.1061	0.1701	<i>0.815</i>
Adaptive TS (ATS)	0.1061	0.1017	0.1683	0.816
Platt Scaling (1D)	0.0740	0.0603	0.1588	<i>0.815</i>
Trajectory Platt (5D)	<i>0.0780</i>	<i>0.0819</i>	<i>0.1600</i>	0.812
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	<i>0.0696</i>	<i>0.0627</i>	0.1259	0.828
Temperature Scaling (TS)	0.0810	0.0671	0.1290	0.828
Adaptive TS (ATS)	0.0690	0.0617	<i>0.1271</i>	0.828
Platt Scaling (1D)	0.0940	0.0946	0.1368	0.828
Trajectory Platt (5D)	0.0960	0.0967	0.1379	<i>0.826</i>
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	0.2045	0.1988	0.2311	0.793
Temperature Scaling (TS)	0.1908	0.1862	0.2218	0.793
Adaptive TS (ATS)	0.1874	0.1858	0.2221	0.793
Platt Scaling (1D)	0.1397	0.1367	0.2034	0.793
Trajectory Platt (5D)	<i>0.1428</i>	<i>0.1434</i>	<i>0.2039</i>	<i>0.792</i>

Table 24: **Temperature Transfer Robustness on vizwiz-vqa (M3-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	0.3066	0.3066	0.2435	0.797
Temperature Scaling (TS)	0.2941	0.2941	0.2328	0.797
Adaptive TS (ATS)	0.2911	0.2911	0.2312	<i>0.796</i>
Platt Scaling (1D)	<i>0.0627</i>	0.0762	<i>0.1433</i>	0.797
Trajectory Platt (5D)	0.0603	<i>0.0817</i>	0.1418	0.795
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	0.6287	0.6287	0.5606	<i>0.756</i>
Temperature Scaling (TS)	0.5041	0.5041	0.4059	<i>0.756</i>
Adaptive TS (ATS)	0.5112	0.5112	0.4144	0.750
Platt Scaling (1D)	<i>0.3687</i>	<i>0.3687</i>	<i>0.3228</i>	<i>0.756</i>
Trajectory Platt (5D)	0.3279	0.3271	0.2960	0.759
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	0.5106	0.5106	0.4138	<i>0.770</i>
Temperature Scaling (TS)	0.4104	0.4104	0.3059	<i>0.770</i>
Adaptive TS (ATS)	0.4142	0.4142	0.3106	0.765
Platt Scaling (1D)	<i>0.1723</i>	<i>0.1723</i>	<i>0.1811</i>	<i>0.770</i>
Trajectory Platt (5D)	0.1353	0.1451	0.1646	0.789
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	0.3569	0.3569	0.2530	0.837
Temperature Scaling (TS)	0.3371	0.3371	0.2374	0.837
Adaptive TS (ATS)	0.3354	0.3354	0.2365	<i>0.834</i>
Platt Scaling (1D)	0.0625	<i>0.0608</i>	<i>0.1182</i>	0.837
Trajectory Platt (5D)	<i>0.0671</i>	0.0606	0.1148	0.837
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	0.2445	0.2445	0.1509	<i>0.884</i>
Temperature Scaling (TS)	0.3184	0.3170	0.1971	<i>0.884</i>
Adaptive TS (ATS)	0.3090	0.3081	0.1904	0.882
Platt Scaling (1D)	<i>0.0708</i>	0.0642	<i>0.0916</i>	<i>0.884</i>
Trajectory Platt (5D)	0.0646	<i>0.0761</i>	0.0905	0.890
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	0.4095	0.4095	0.3244	<i>0.809</i>
Temperature Scaling (TS)	0.3728	0.3725	0.2758	<i>0.809</i>
Adaptive TS (ATS)	0.3722	0.3720	0.2766	0.805
Platt Scaling (1D)	<i>0.1474</i>	<i>0.1484</i>	<i>0.1714</i>	<i>0.809</i>
Trajectory Platt (5D)	0.1311	0.1381	0.1615	0.814

Table 25: **Temperature Transfer Robustness on vizwiz-vqa (MQT-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.4200	0.4200	0.4200	0.614
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.0862	0.1218	0.2477	0.614
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.0864	0.1315	0.2476	<i>0.625</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.0563	0.0770	<i>0.2370</i>	0.614
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>0.0861</i>	<i>0.1115</i>	0.2289	0.661

Table 26: **Calibration Benchmark on ai2d (M3-LLaVA 7B)**. Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.2715	0.2692	0.2887	<i>0.746</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.1331	0.1310	0.2197	<i>0.746</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.1338	0.1408	0.2195	0.747
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.0592	<i>0.0825</i>	0.2015	<i>0.746</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>0.0627</i>	0.0653	<i>0.2035</i>	0.739

Table 27: **Calibration Benchmark on ai2d (MQT-LLaVA 7B)**. Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

3 Group 1: Per-Dataset In-Domain Calibration Benchmarks

Individual benchmark tables for all vision-language datasets under the 4-method, 4-metric schema.

3.1 Benchmark: AI2D

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.2566	0.2540	0.3194	0.599
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	<i>0.0445</i>	0.1796	0.2480	0.599
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.0429	<i>0.1694</i>	0.2477	<i>0.601</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.0684	0.1767	<i>0.2463</i>	0.599
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.0526	0.0745	0.2240	0.691

Table 28: **Calibration Benchmark on avqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.2788	0.2800	0.3385	0.602
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.0656	0.1328	0.2491	0.602
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.0662	<i>0.1286</i>	0.2488	<i>0.604</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.0118	0.1366	<i>0.2449</i>	0.602
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>0.0502</i>	0.0654	0.2332	0.654

Table 29: **Calibration Benchmark on avqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

3.2 Benchmark: Adversarial VQA

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.7103	0.7103	0.6151	<i>0.807</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.4447	0.4447	0.2782	<i>0.807</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.4445	0.4445	0.2781	<i>0.807</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.0323	<i>0.0362</i>	<i>0.0755</i>	<i>0.807</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>0.0331</i>	0.0288	0.0752	0.818

Table 30: **Calibration Benchmark on chartqa (M3-LLaVA 7B).** Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.7130	0.7130	0.6483	0.491
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.4488	0.4488	0.2779	0.491
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.4487	0.4487	0.2777	0.491
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>0.0306</i>	<i>0.0427</i>	0.0762	<i>0.509</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.0244	0.0392	<i>0.0765</i>	0.603

Table 31: **Calibration Benchmark on chartqa (MQT-LLaVA 7B).** Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

3.3 Benchmark: CHARTQA

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.7457	0.7457	0.6430	<i>0.845</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.4889	0.4889	0.2796	<i>0.845</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.4887	0.4887	0.2794	<i>0.845</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>0.0262</i>	<i>0.0252</i>	<i>0.0405</i>	<i>0.845</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.0187	0.0181	0.0377	0.888

Table 32: **Calibration Benchmark on docvqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.5330	0.5330	0.4162	<i>0.517</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.4707	0.4707	0.2584	<i>0.517</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.4706	0.4706	0.2583	<i>0.517</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.0009	<i>0.0367</i>	0.0362	<i>0.517</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>0.0108</i>	0.0206	<i>0.0364</i>	0.594

Table 33: **Calibration Benchmark on docvqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

3.4 Benchmark: DOCVQA

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.3255	0.3255	0.3223	0.683
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.0841	0.1085	0.2082	0.683
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	<i>0.0747</i>	0.1125	0.2053	0.709
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.0590	<i>0.0689</i>	<i>0.1991</i>	0.683
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.0761	0.0650	0.1978	<i>0.707</i>

Table 34: **Calibration Benchmark on gqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.1364	0.1019	0.1865	0.757
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	<i>0.0921</i>	0.1061	0.1860	0.757
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.0758	0.1199	<i>0.1857</i>	<i>0.759</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.0956	<i>0.1060</i>	0.1861	0.757
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.1381	0.1466	0.1854	0.763

Table 35: **Calibration Benchmark on gqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

3.5 Benchmark: GQA

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.8532	0.8532	0.7761	0.333
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.5400	0.5400	0.3006	0.333
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.5398	0.5398	0.3004	0.333
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>0.0003</i>	<i>0.0122</i>	0.0074	0.667
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.0001	0.0120	<i>0.0075</i>	<i>0.501</i>

Table 36: **Calibration Benchmark on infographicvqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.7551	0.7551	0.6535	<i>0.769</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.5263	0.5263	0.2826	<i>0.769</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.5261	0.5261	0.2825	<i>0.769</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.0028	<i>0.0111</i>	<i>0.0050</i>	<i>0.769</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>0.0030</i>	0.0108	0.0049	0.930

Table 37: **Calibration Benchmark on infographicvqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

3.6 Benchmark: INFOGRAPHICVQA

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.7316	0.7316	0.7314	<i>0.477</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.3471	0.3471	0.3160	<i>0.477</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.3466	0.3466	0.3156	<i>0.477</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.0071	<i>0.1189</i>	<i>0.1958</i>	<i>0.477</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>0.0415</i>	0.0940	0.1914	0.601

Table 38: **Calibration Benchmark on lego-puzzles (M3-LLaVA 7B)**. Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.3460	0.3380	0.3474	0.510
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.2368	0.2368	0.2539	0.510
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.2368	0.2368	0.2539	<i>0.511</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.0060	<i>0.1243</i>	<i>0.1968</i>	0.490
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>0.0781</i>	0.0867	0.1924	0.593

Table 39: **Calibration Benchmark on lego-puzzles (MQT-LLaVA 7B)**. Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

3.7 Benchmark: LEGOPUZZLES

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.2700	0.2700	0.2700	0.608
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.0643	0.0779	0.1956	0.608
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.0286	0.0737	<i>0.1911</i>	<i>0.655</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>0.0460</i>	<i>0.0694</i>	0.1970	0.608
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.0662	0.0654	0.1890	0.660

Table 40: **Calibration Benchmark on mmbench (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.1538	0.1527	0.1866	<i>0.828</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.0981	0.1024	0.1560	<i>0.828</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	<i>0.0905</i>	0.1014	0.1558	0.829
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.0811	<i>0.0838</i>	0.1503	<i>0.828</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.1042	0.0751	<i>0.1543</i>	0.819

Table 41: **Calibration Benchmark on mmbench (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

3.8 Benchmark: MMBENCH

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.7876	0.7876	0.6889	0.639
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.5221	0.5221	0.2908	0.639
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.5219	0.5219	0.2906	0.639
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>0.0181</i>	0.0344	<i>0.0205</i>	0.639
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.0168	<i>0.0356</i>	0.0198	<i>0.623</i>

Table 42: **Calibration Benchmark on mmmu (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.5224	0.5224	0.3669	<i>0.315</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.4932	0.4932	0.2556	<i>0.315</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.4931	0.4931	0.2555	<i>0.315</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.0016	<i>0.0214</i>	0.0115	<i>0.315</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>0.0023</i>	0.0204	<i>0.0116</i>	0.724

Table 43: **Calibration Benchmark on mmmu (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

3.9 Benchmark: MMMU

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.0345	0.0440	<i>0.0882</i>	0.739
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.0427	0.0437	0.0883	0.739
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.0442	0.0427	0.0879	<i>0.750</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>0.0419</i>	<i>0.0432</i>	<i>0.0882</i>	0.739
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.0482	0.0662	0.0887	0.751

Table 44: **Calibration Benchmark on pope (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.0401	0.0430	0.0903	0.876
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.0352	0.0493	<i>0.0902</i>	0.876
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.0375	0.0490	<i>0.0902</i>	<i>0.874</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.0397	0.0464	0.0893	0.876
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>0.0355</i>	<i>0.0448</i>	0.0903	0.873

Table 45: **Calibration Benchmark on pope (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

3.10 Benchmark: POPE

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.2216	0.2193	0.2368	0.807
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.1310	0.1268	0.1924	0.807
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.1298	0.1300	0.1918	<i>0.808</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.0480	0.0499	<i>0.1732</i>	0.807
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>0.0517</i>	<i>0.0615</i>	0.1719	0.813

Table 46: **Calibration Benchmark on scienceqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.2365	0.2317	0.2579	0.760
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.0932	0.1023	0.1975	0.760
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.0873	0.0938	0.1966	<i>0.761</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>0.0799</i>	<i>0.0691</i>	<i>0.1916</i>	0.760
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.0646	0.0572	0.1860	0.769

Table 47: **Calibration Benchmark on scienceqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

3.11 Benchmark: SCIENCEQA

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.3225	0.3225	0.3225	0.483
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	<i>0.0066</i>	<i>0.0525</i>	0.2185	0.483
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.0087	0.0860	<i>0.2180</i>	<i>0.554</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.0048	0.0520	0.2184	0.483
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.0394	0.0768	0.2071	0.642

Table 48: **Calibration Benchmark on seedbench (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.1856	0.1774	0.2297	0.733
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.0668	0.1044	0.2015	0.733
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.0604	<i>0.0965</i>	0.2009	<i>0.736</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.0951	0.0862	<i>0.1986</i>	0.733
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>0.0666</i>	0.0967	0.1957	0.741

Table 49: **Calibration Benchmark on seedbench (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

3.12 Benchmark: SEEDBENCH

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.1050	0.1107	0.1714	0.839
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.1038	0.1114	0.1712	0.839
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.1015	0.1131	0.1710	0.839
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>0.0847</i>	<i>0.0822</i>	0.1615	0.839
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.0578	0.0535	<i>0.1628</i>	<i>0.835</i>

Table 50: **Calibration Benchmark on textvqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.3227	0.3191	0.3286	0.684
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.1821	0.1832	0.2503	0.684
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.1836	0.1836	0.2500	<i>0.689</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>0.0895</i>	<i>0.1046</i>	<i>0.2148</i>	0.684
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.0525	0.0776	0.1948	0.719

Table 51: **Calibration Benchmark on textvqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

3.13 Benchmark: TEXTVQA

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.1597	0.1597	0.2071	<i>0.793</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.1509	0.1509	0.2065	<i>0.793</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.1487	0.1487	<i>0.2058</i>	<i>0.793</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>0.0664</i>	0.0580	0.1828	<i>0.793</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.0638	<i>0.0657</i>	0.1828	0.794

Table 52: **Calibration Benchmark on vizwiz-vqa (M3-LLaVA 7B)**. Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.3066	0.3066	0.2435	0.797
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.2941	0.2941	0.2328	0.797
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.2911	0.2911	0.2312	<i>0.796</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>0.0627</i>	0.0762	<i>0.1433</i>	0.797
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.0603	<i>0.0817</i>	0.1418	0.795

Table 53: **Calibration Benchmark on vizwiz-vqa (MQT-LLaVA 7B)**. Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

3.14 Benchmark: VIZWIZVQA

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	<i>0.1159</i>	<i>0.1050</i>	0.1732	0.814
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.1698	0.1680	0.1812	0.814
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.1408	0.1384	<i>0.1730</i>	<i>0.813</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.1476	0.1404	0.1798	0.814
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.0587	0.0688	0.1709	0.773

Table 54: **Calibration Benchmark on vllm-safety (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	<i>0.0963</i>	<i>0.0974</i>	0.1966	<i>0.763</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.1420	0.1442	0.2054	<i>0.763</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.1213	0.1319	0.1966	0.768
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.1134	0.1129	<i>0.1937</i>	<i>0.763</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.0568	0.0658	0.1921	0.733

Table 55: **Calibration Benchmark on vllm-safety (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

3.15 Benchmark: VLLM-SAFETY

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	0.0864	0.0723	0.1387	<i>0.772</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.0940	0.0858	0.1417	<i>0.772</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.0839	0.0752	<i>0.1381</i>	0.788
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.0552	<i>0.0697</i>	0.1391	<i>0.772</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>0.0686</i>	0.0650	0.1380	0.788

Table 56: **Calibration Benchmark on vqav2 (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE ↓	Ada-ECE ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	<i>0.1049</i>	<i>0.1031</i>	0.1713	0.641
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	0.1387	0.1365	0.1789	0.641
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.1236	0.1203	0.1753	<i>0.658</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.1054	0.1094	<i>0.1712</i>	0.641
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.0716	0.0764	0.1629	0.739

Table 57: **Calibration Benchmark on vqav2 (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

3.16 Benchmark: VQAV2

Cardinality (k)	Subsets $\binom{5}{k}$	Optimal Formula	Macro Ada-ECE (Best) ↓	Macro Ada-ECE (Mean ± Std)
M3-LLaVA (7B)				
$k = 1$	5	x_5	0.0323	0.0430 ± 0.0108
$k = 2$	10	$x_2 + x_4$	<i>0.0314</i>	0.0435 ± 0.0078
$k = 3$	10	$x_2 + x_4 + x_5$	0.0290	0.0416 ± 0.0057
$k = 4$	5	$x_1 + x_3 + x_4 + x_5$	0.0356	0.0391 ± 0.0024
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0365	0.0365 ± 0.0000
MQT-LLaVA (7B)				
$k = 1$	5	x_5	0.0297	0.0461 ± 0.0154
$k = 2$	10	$x_4 + x_5$	<i>0.0268</i>	0.0448 ± 0.0139
$k = 3$	10	$x_2 + x_4 + x_5$	0.0235	0.0397 ± 0.0091
$k = 4$	5	$x_1 + x_2 + x_3 + x_4$	0.0341	0.0368 ± 0.0029
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0344	0.0344 ± 0.0000

Table 58: **Combinatorial 5D Trajectory Feature Cardinality Progression** ($k \in \{1 \dots 5\}$). Macro-averaged out-of-fold calibration error and discrimination across Core 5 benchmarks for M3-LLaVA and MQT-LLaVA (7B). Evaluates all $\sum_{k=1}^5 \binom{5}{k} = 31$ feature subsets. **Bold**: Rank 1, *Italic*: Rank 2 within each architecture.

4 Group 8: Combinatorial 5D Trajectory Feature Ablation Study

Exhaustive 5-fold cross-validation combinatorial ablation across all $\sum_{k=1}^5 \binom{5}{k} = 31$ feature subsets of canonical 5D trajectory features across all 14 vision-language benchmarks on M3-LLaVA (7B) and MQT-LLaVA (7B).

4.1 Table 4: Macro Cardinality Progression ($k \in \{1 \dots 5\}$)

Dropped Feature	Functional Role	M3-LLaVA (7B)		MQT-LLaVA
		Δ Ada-ECE	Δ AUROC	Δ Ada-ECE
x_1 (Final Logit)	Base Fine-Scale Log-Odds Anchor ($\ell_{576/256}$)	<i>+0.0038</i>	-0.094	+0.0075
x_2 (Discrete Answer Stability)	Categorical Scale Consistency	-0.0009	-0.001	+0.0014
x_3 (Scale Entropy Slope)	Predictive Entropy Decay Dynamics	+0.0063	<i>-0.002</i>	<i>+0.0035</i>
x_4 (Answer Flip Frequency)	Argmax Scale Transition Rate	+0.0014	-0.001	+0.0000
x_5 (Monotonicity Count)	Directional Monotonicity Trend	+0.0023	+0.000	-0.0003

Table 59: **Leave-One-Out (LOO) Feature Degradation Sensitivity across Core 5 Benchmarks.** Marginal change in calibration error (Δ Ada-ECE) and discrimination (Δ AUROC) when dropping individual feature x_j from Trajectory Platt (5D). **Bold:** Rank 1, *Italic:* Rank 2 degradation impact.

4.2 Table 5: Leave-One-Out (LOO) Feature Degradation Sensitivity

Benchmark Dataset	1D Platt ($\{x_1\}$)	Best-2D	Best-3D	Best-4D	Full 5D (Ours)
M3-LLaVA (7B)					
AI2D	0.0418	0.0311	<i>0.0335</i>	0.0440	0.0620
ChartQA	0.0149	0.0124	<i>0.0129</i>	0.0139	0.0153
DocVQA	0.0115	<i>0.0085</i>	0.0088	0.0083	0.0121
GQA	0.0411	<i>0.0255</i>	0.0243	0.0261	0.0332
InfoVQA	0.0053	0.0044	<i>0.0047</i>	0.0056	0.0074
LegoPuzzles	0.0949	0.0459	<i>0.0357</i>	0.0413	0.0345
MMBench	0.0419	0.0256	<i>0.0271</i>	0.0355	0.0420
MMMU	0.0131	0.0092	<i>0.0098</i>	0.0122	0.0155
POPE	0.0137	<i>0.0116</i>	0.0104	0.0135	0.0185
ScienceQA	0.0407	0.0208	<i>0.0292</i>	0.0370	0.0351
SEED-Bench	0.0274	<i>0.0259</i>	0.0269	0.0263	0.0230
TextVQA	0.0432	<i>0.0226</i>	0.0166	0.0359	0.0376
VizWiz-VQA	0.0259	0.0172	0.0202	<i>0.0188</i>	0.0204
VQAv2	0.0764	0.0204	<i>0.0206</i>	0.0238	0.0272
Macro Mean	0.0351	0.0201	0.0201	<i>0.0244</i>	0.0274
MQT-LLaVA (7B)					
AI2D	0.0362	0.0152	<i>0.0176</i>	0.0294	0.0371
ChartQA	0.0454	0.0165	<i>0.0123</i>	0.0122	0.0179
DocVQA	0.0231	<i>0.0083</i>	0.0077	<i>0.0083</i>	0.0118
GQA	0.0525	0.0287	<i>0.0335</i>	0.0353	0.0544
InfoVQA	0.0065	0.0063	0.0054	0.0046	<i>0.0051</i>
LegoPuzzles	0.0899	0.0395	0.0303	<i>0.0383</i>	0.0520
MMBench	0.0407	0.0321	<i>0.0391</i>	0.0453	0.0490
MMMU	0.0128	0.0098	0.0079	<i>0.0090</i>	0.0116
POPE	0.0236	0.0198	<i>0.0175</i>	0.0171	0.0210
ScienceQA	0.0511	0.0329	0.0282	<i>0.0310</i>	0.0404
SEED-Bench	0.0303	0.0227	<i>0.0299</i>	0.0338	0.0530
TextVQA	0.0997	0.0223	<i>0.0225</i>	0.0386	0.0345
VizWiz-VQA	0.0446	0.0092	<i>0.0237</i>	0.0302	0.0305
VQAv2	0.0880	0.0232	0.0256	<i>0.0255</i>	0.0296
Macro Mean	0.0460	0.0205	<i>0.0215</i>	0.0256	0.0320

Table 60: **Exhaustive 14-Benchmark Progression Grid: Adaptive ECE ($\downarrow$)**. Per-dataset comparison across 1D Platt ($\{x_1\}$), Optimal 2D, Optimal 3D, Optimal 4D, and Full Trajectory Platt 5D. Option A ranking: **Bold** Rank 1, *Italic* Rank 2 across each benchmark row.

4.3 Table 6: Exhaustive 14-Benchmark Adaptive ECE Progression Grid

Cardinality (k)	Subsets	Optimal Formula	Ada-ECE ↓	Mean ± Std	AUROC ↑	Brier ↓
M3-LLaVA (7B)						
$k = 1$	5	x_5	0.0297	0.0504 ± 0.0192	0.583	0.2371
$k = 2$	10	$x_4 + x_5$	<i>0.0226</i>	0.0501 ± 0.0144	<i>0.731</i>	0.2039
$k = 3$	10	$x_2 + x_4 + x_5$	0.0166	0.0425 ± 0.0111	0.730	0.2040
$k = 4$	5	$x_1 + x_3 + x_4 + x_5$	0.0359	0.0402 ± 0.0027	0.842	<i>0.1595</i>
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0376	0.0376 ± 0.0000	0.842	0.1593
MQT-LLaVA (7B)						
$k = 1$	5	x_5	0.0223	0.0527 ± 0.0307	0.607	0.2197
$k = 2$	10	$x_2 + x_4$	0.0223	0.0456 ± 0.0149	0.633	0.2163
$k = 3$	10	$x_2 + x_4 + x_5$	<i>0.0225</i>	0.0427 ± 0.0115	0.682	0.2070
$k = 4$	5	$x_1 + x_2 + x_4 + x_5$	0.0386	0.0448 ± 0.0054	<i>0.738</i>	<i>0.1926</i>
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0345	0.0345 ± 0.0000	0.747	0.1851

Table 61: **Combinatorial 5D Trajectory Feature Cardinality Progression on TextVQA** ($k \in \{1 \dots 5\}$). Out-of-fold calibration error and discrimination across feature cardinality levels for M3-LLaVA and MQT-LLaVA (7B). **Bold**: Rank 1, *Italic*: Rank 2 within each architecture.

4.4 Table 7: Cardinality Progression on TextVQA

Cardinality (k)	Subsets	Optimal Formula	Ada-ECE ↓	Mean ± Std	AUROC ↑	Brier ↓
M3-LLaVA (7B)						
$k = 1$	5	x_5	0.0271	0.0411 ± 0.0202	0.580	0.1634
$k = 2$	10	$x_4 + x_5$	0.0204	0.0408 ± 0.0169	0.710	0.1481
$k = 3$	10	$x_2 + x_4 + x_5$	<i>0.0206</i>	0.0361 ± 0.0111	0.714	0.1464
$k = 4$	5	$x_2 + x_3 + x_4 + x_5$	0.0238	0.0316 ± 0.0059	<i>0.752</i>	<i>0.1431</i>
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0272	0.0272 ± 0.0000	0.788	0.1362
MQT-LLaVA (7B)						
$k = 1$	5	x_4	0.0259	0.0430 ± 0.0257	0.661	0.1633
$k = 2$	10	$x_4 + x_5$	0.0232	0.0385 ± 0.0140	0.703	0.1611
$k = 3$	10	$x_2 + x_4 + x_5$	0.0256	0.0327 ± 0.0062	0.700	0.1614
$k = 4$	5	$x_1 + x_2 + x_4 + x_5$	<i>0.0255</i>	0.0285 ± 0.0019	<i>0.729</i>	<i>0.1558</i>
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0296	0.0296 ± 0.0000	0.771	0.1511

Table 62: **Combinatorial 5D Trajectory Feature Cardinality Progression on VQAv2** ($k \in \{1 \dots 5\}$). Out-of-fold calibration error and discrimination across feature cardinality levels for M3-LLaVA and MQT-LLaVA (7B). **Bold**: Rank 1, *Italic*: Rank 2 within each architecture.

4.5 Table 8: Cardinality Progression on VQAv2

Cardinality (k)	Subsets	Optimal Formula	Ada-ECE ↓	Mean ± Std	AUROC ↑	Brier ↓
M3-LLaVA (7B)						
$k = 1$	5	x_5	0.0411	0.0581 ± 0.0216	0.516	0.1971
$k = 2$	10	$x_2 + x_4$	0.0459	0.0589 ± 0.0151	0.494	0.1981
$k = 3$	10	$x_3 + x_4 + x_5$	<i>0.0357</i>	0.0568 ± 0.0186	0.589	0.1927
$k = 4$	5	$x_1 + x_2 + x_3 + x_5$	0.0413	0.0495 ± 0.0106	<i>0.583</i>	0.1927
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0345	0.0345 ± 0.0000	0.579	<i>0.1932</i>
MQT-LLaVA (7B)						
$k = 1$	5	x_2	<i>0.0337</i>	0.0549 ± 0.0212	0.485	0.1982
$k = 2$	10	$x_2 + x_4$	0.0395	0.0623 ± 0.0260	0.485	0.2002
$k = 3$	10	$x_1 + x_3 + x_4$	0.0303	0.0535 ± 0.0224	0.613	0.1913
$k = 4$	5	$x_1 + x_2 + x_3 + x_4$	0.0383	0.0536 ± 0.0172	0.607	0.1934
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0520	0.0520 ± 0.0000	<i>0.611</i>	<i>0.1929</i>

Table 63: **Combinatorial 5D Trajectory Feature Cardinality Progression on LegoPuzzles** ($k \in \{1 \dots 5\}$). Out-of-fold calibration error and discrimination across feature cardinality levels for M3-LLaVA and MQT-LLaVA (7B). **Bold**: Rank 1, *Italic*: Rank 2 within each architecture.

4.6 Table 9: Cardinality Progression on LegoPuzzles